

Building Seismic-Resilience Credit Model (MKM-SEIS-001)

Wolfram Oracle Benchmark: the Credit Computed Exactly

MKM Research Labs

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Contents

1 Executive Summary **3**

2 Model Purpose and Scope **3**

2.1 What is benchmarked 3

2.2 What is not 4

3 Mathematical Framework **4**

4 Benchmark Method **4**

5 Validation and Backtesting **5**

5.1 The model document's claims 5

5.2 How much the tables can say 6

5.3 Every printed cell 6

6 Model Limitations and Known Weaknesses **8**

7 Document History **8**

1 Executive Summary

This document benchmarks the Building Seismic-Resilience Credit Model (MKM-SEIS-001), which prices a commercial asset's seismic resilience as a no-collapse rate, an annual loss frequency and a full-collapse (DS3) spread for the Commercial PRS Pricer. The model estimates all three by Monte Carlo. This benchmark computes them *exactly*: with the source-to-site distance fixed, an event's damage state depends on its magnitude, the GMPE residual and one fragility uniform, and the last two integrate out in closed form, leaving one integral over the truncated Gutenberg–Richter law per damage state.

The model is right, and its tables are honest. Model C, the GMPE median and the occurrence rate match the reference to 6.2×10^{-16}. The code, run at the document's own draw count and seed, reproduces 131 of the 131 figures printed in its Tables 3–10, and every printed figure lies within 1.94 standard errors of the exact credit (Table 5). All six acceptance gates of Table 2 hold on the exact credit, without the 0.7 bp tolerance the generator allows for noise. Wolfram's integrals agree with the 50-digit reference to 4.6×10^{-31}.

Findings (Table 2):

1. **The tables are too noisy to read cell by cell.** At 20,000 draws one collapse is 0.5 bp, and a 1.03 bp spread carries a 0.72 bp standard error (70%) — so §9.6's "stability is adequate at $n_{\text{sim}} = 20{,}000$" does not hold. 4 adjacent pairs in the tables run against the exact order, among them Table 9's Very-high row, where B prints above NR: all are sampling noise (Table 3).

2. **The production spread is noisy too.** Production runs 10,000 draws; the largest exact spread, 31.7 bp, then carries a 5.6 bp standard error. The exact credit takes under a second per asset.

3. **Table 18's "Mean L" are medians.** The loss of DS0–DS2 is a lognormal with that *median*, clipped (not truncated) at 1, so its mean is higher: DS2 averages 0.379, not 0.35, before the post-earthquake-fire cascade adds more.

4. **The GMPE is not specified.** §3.3 and Appendix B name a Boore–Atkinson-style proxy; the coded form, with magnitude-dependent geometric spreading, appears only in the code.

Evidence. 7 checks, outcome: all pass (Table 1).

- Wolfram: 15.0.0 for Mac OS X ARM (64-bit) (May 26, 2026) via live kernel session (wolframclient).
- Code under test, at commit `7bbb63078a59`.
- Numerics: numpy 1.26.4, scipy 1.15.1, mpmath 1.4.1.
- Generated 2026-10-28 by `python -m tools.wolfram_oracle seis001`.

2 Model Purpose and Scope

2.1 What is benchmarked

In `src/models/seismic/`: occurrence and the fault geometry (Model A), the GMPE (Model B), the BRI modifiers (Model C) and the damage, loss and cascade of Model D, end to end through

`simulate_asset_seismic`, each cell built as the document's sensitivity generator built it: a 2 km east–west fault with the asset due south of its midpoint.

2.2 What is not

Table 11 prices the Halong portfolio from catchment data not available to this run. Assets with a long fault and an off-centre position, where R_{JB} varies with the rupture, are outside the generator's cases and this benchmark's. The functionality index and PML outputs do not enter the credit. Every parameter is an engineering-judgement seed (SEIS-L1). This addresses validation question Q8, not Q9.

3 Mathematical Framework

With M truncated Gutenberg–Richter on $[m_{\min}, m_{\max}]$ with slope b, the coded GMPE median $\mu(M, R, V_{S30})$, the soft-soil factor F_a, residual sigma σ, and Model-C-adjusted medians θ_i with dispersion β, one uniform against the nested fragility curves gives

$$P(\mathrm{DS} \geq i \mid M) = \Phi\left(\frac{\ln \mu(M) + \ln F_a - \ln \theta_i}{\sqrt{\sigma^2 + \beta^2}}\right), \qquad p_i = \int_{m_{\min}}^{m_{\max}} P(\mathrm{DS} \geq i \mid m)\, f_M(m)\, \mathrm{d}m. \tag{1}$$

With $q = 1 - e^{-\lambda h}$ the probability a draw has an event, the spread is $10^4 q\, p_3$ and the no-collapse rate $1 - p_3$. The loss of DS0–DS2 is $\min(\tilde{m} e^{sZ}, 1)$, whose mean is $\tilde{m} e^{s^2/2}\Phi(-\ln \tilde{m}/s - s) + \Phi(\ln \tilde{m}/s)$; the cascade replaces it with the larger of it and the fire loss, whose mean is $\int_0^1 (1 - F_S(x) F_F(x))\mathrm{d}x$; the loss frequency is $q \sum_k P(\mathrm{DS} = k)\, E[L \mid k]$.

4 Benchmark Method

The integrals are evaluated with `mpmath` at 50 digits, from the document and the configuration rather than the code; the GMPE takes the form the code documents, the document giving none. Wolfram evaluates the damage-state integrals independently with `NIntegrate`, for each of the 66 distinct assets behind the 89 printed cells. The model runs each cell at 20,000 draws with seed 20260608, as the generator did, and each figure is judged in standard errors of the exact value: the no-collapse rate's on the expected number of events, the loss frequency's from the exact second moment, and the median PGA's from the density at the median.

5 Validation and Backtesting

Table 1: Benchmark checks

Check	Result	Limit	Outcome
Model C medians, GMPE median and λ_i against the reference	6.2×10^{-16}	1.0×10^{-12}	pass
Every event's R_{JB} is the asset's offset (the generator's geometry)	6.5×10^{-14}	1.0×10^{-12}	pass
Median PGA: model's Monte Carlo against the exact credit, worst $\lvert z \rvert$	1.15	4	pass
No-collapse rate: model's Monte Carlo against the exact credit, worst $\lvert z \rvert$	1.76	4	pass
Loss frequency: model's Monte Carlo against the exact credit, worst $\lvert z \rvert$	1.92	4	pass
Seismic spread: model's Monte Carlo against the exact credit, worst $\lvert z \rvert$	1.94	4	pass
Wolfram's integrals over magnitude agree with the reference	4.6×10^{-31}	1.0×10^{-20}	pass

5.1 The model document's claims

Table 2: The model document's claims, tested

Claim	Stated	Benchmark finds	Outcome
§7, Tables 3–10	the tables come from the running model	the code reproduces 131 of 131 printed figures at the generator's seed	met
§7, Tables 3–10	the figures are the model's credit	every printed figure within 1.94 standard errors of the exact value	met
Table 2	six acceptance gates pass	all hold on the exact credit, without the harness's 0.7 bp tolerance; the floor is 0.0002 bp	met
§7	monotone in distance, zone, soil and resilience	4 adjacent printed pairs run against the exact order; all are sampling noise	finding
§9.6	stability is adequate at 20,000 draws	a 1.03 bp spread carries a 0.72 bp standard error (70%)	**not met**
Table 18	mean loss 0.01, 0.10, 0.35 (DS0–DS2), truncated lognormal	the figures are medians of lognormals clipped at 1: means 0.0113, 0.1133, 0.3785	**not met**
§3.3, App. B	a Boore–Atkinson-style GMPE	the coded form, with magnitude-dependent spreading $(c_3 + c_5(M - m_{\text{ref}})) \ln R$, is not stated	finding

5.2 How much the tables can say

Table 3: The largest spreads and their standard errors at 20,000 draws (the tables) and 10,000 (production)

Cell	Exact (bp)	Std error, tables (bp)	Std error, production (bp)
T6/Very high / Soft (D) no GS02	31.68	3.97	5.62
T6/Very high / Soft (D) GS02	19.88	3.15	4.45
T4/2 / Masonry URM	14.76	2.71	3.84
T6/High / Soft (D) no GS02	12.02	2.45	3.47
T3/2 km	8.16	2.02	2.86
T4/2 / RC pre-code	8.16	2.02	2.86
T10/2 / NR	8.16	2.02	2.86
T5/Very high	7.58	1.95	2.75

Table 4: Table 3's median PGA: printed against the exact median of the event PGA

Cell	R_{JB} (km)	Printed (g)	Exact (g)	$\lvert z\rvert$
2 km	2.004	0.1157	0.1090	0.78
5 km	5.009	0.0835	0.0783	0.83
10 km	10.018	0.0520	0.0485	0.89
20 km	20.035	0.0291	0.0270	0.98
50 km	50.088	0.0129	0.0118	1.16
100 km	100.176	0.0068	0.0062	1.09

5.3 Every printed cell

Table 5: Every printed cell against the exact credit: no-collapse (%), loss frequency (%), spread (bp), printed then exact; $\lvert z\rvert$ in standard errors at 20,000 draws

T	Cell	No-collapse		Loss		Spread		$\lvert z\rvert$
3	2 km	93.1	89.8	0.1228	0.1612	5.00	8.16	1.64
3	5 km	94.4	94.0	0.0954	0.1102	4.00	4.80	0.83
3	10 km	97.9	97.4	0.0526	0.0609	1.50	2.07	0.89
3	20 km	100.0	99.2	0.0151	0.0295	0.00	0.67	1.90
3	50 km	100.0	99.9	0.0088	0.0137	0.00	0.11	1.42
3	100 km	100.0	100.0	0.0083	0.0103	0.00	0.02	1.10
4	2 / RC pre-code	–	–	–	–	5.00	8.16	1.57
4	2 / Masonry URM	–	–	–	–	9.50	14.76	1.94
4	2 / RC modern	–	–	–	–	3.50	3.90	0.29
4	2 / Steel MRF	–	–	–	–	2.50	3.00	0.41
4	10 / RC pre-code	–	–	–	–	1.50	2.07	0.56
4	10 / Masonry URM	–	–	–	–	3.00	4.35	0.91
4	10 / RC modern	–	–	–	–	0.00	0.85	1.30
4	10 / Steel MRF	–	–	–	–	0.00	0.62	1.12
4	20 / RC pre-code	–	–	–	–	0.00	0.67	1.15
4	20 / Masonry URM	–	–	–	–	0.50	1.54	1.18
4	20 / RC modern	–	–	–	–	0.00	0.24	0.70
4	20 / Steel MRF	–	–	–	–	0.00	0.17	0.59
4	50 / RC pre-code	–	–	–	–	0.00	0.11	0.47

Table 5: Every printed cell against the exact credit: no-collapse (%), loss frequency (%), spread (bp), printed then exact; $|z|$ in standard errors at 20,000 draws

T	Cell	No-collapse		Loss		Spread		$\lvert z\rvert$
4	50 / Masonry URM	–	–	–	–	0.00	0.30	0.77
4	50 / RC modern	–	–	–	–	0.00	0.03	0.26
4	50 / Steel MRF	–	–	–	–	0.00	0.02	0.21
5	Very low	100.0	99.6	0.0012	0.0015	0.00	0.02	0.20
5	Low	100.0	99.4	0.0061	0.0069	0.00	0.13	0.51
5	Moderate	100.0	99.2	0.0151	0.0295	0.00	0.67	1.90
5	High	99.1	99.1	0.1067	0.1133	2.50	2.78	0.44
5	Very high	99.4	99.0	0.2868	0.2987	4.50	7.58	1.58
6	Low / Rock (A)	–	–	–	–	0.00	0.07	0.38
6	Low / Stiff (B)	–	–	–	–	0.00	0.13	0.51
6	Low / Soft (D) no GS02	–	–	–	–	0.00	0.69	1.18
6	Low / Soft (D) GS02	–	–	–	–	0.50	0.40	0.22
6	Moderate / Rock (A)	–	–	–	–	0.00	0.40	0.89
6	Moderate / Stiff (B)	–	–	–	–	0.00	0.67	1.15
6	Moderate / Soft (D) no GS02	–	–	–	–	2.50	3.10	0.48
6	Moderate / Soft (D) GS02	–	–	–	–	2.50	1.89	0.63
6	High / Rock (A)	–	–	–	–	2.00	1.72	0.30
6	High / Stiff (B)	–	–	–	–	2.50	2.78	0.24
6	High / Soft (D) no GS02	–	–	–	–	9.50	12.02	1.03
6	High / Soft (D) GS02	–	–	–	–	4.00	7.48	1.80
6	Very high / Rock (A)	–	–	–	–	5.00	4.79	0.14
6	Very high / Stiff (B)	–	–	–	–	4.50	7.58	1.58
6	Very high / Soft (D) no GS02	–	–	–	–	33.00	31.68	0.33
6	Very high / Soft (D) GS02	–	–	–	–	21.50	19.88	0.51
7	NR	94.4	94.0	0.0954	0.1102	4.00	4.80	0.80
7	B	97.2	95.3	0.0645	0.0910	2.00	3.75	1.62
7	A	99.3	97.4	0.0445	0.0640	0.50	2.04	1.59
7	AA	100.0	99.0	0.0226	0.0383	0.00	0.77	1.93
8	None	94.4	94.0	–	–	4.00	4.80	0.52
8	GS08	95.8	95.6	–	–	3.00	3.47	0.36
8	GS09	95.8	95.1	–	–	3.00	3.93	0.67
8	GS12	97.2	96.7	–	–	2.00	2.61	0.53
8	GS14	95.1	95.3	–	–	3.50	3.75	0.18
8	GS02	94.4	94.0	–	–	4.00	4.80	0.52
9	Very low / NR	–	–	–	–	0.00	0.02	0.20
9	Very low / B	–	–	–	–	0.00	0.01	0.16
9	Very low / A	–	–	–	–	0.00	0.00	0.10
9	Very low / AA	–	–	–	–	0.00	0.00	0.04
9	Low / NR	–	–	–	–	0.00	0.13	0.51
9	Low / B	–	–	–	–	0.00	0.09	0.43
9	Low / A	–	–	–	–	0.00	0.04	0.28
9	Low / AA	–	–	–	–	0.00	0.01	0.14
9	Moderate / NR	–	–	–	–	0.00	0.67	1.15
9	Moderate / B	–	–	–	–	0.00	0.49	0.99
9	Moderate / A	–	–	–	–	0.00	0.23	0.67
9	Moderate / AA	–	–	–	–	0.00	0.07	0.37
9	High / NR	–	–	–	–	2.50	2.78	0.24
9	High / B	–	–	–	–	2.00	2.08	0.08

Table 5: Every printed cell against the exact credit: no-collapse (%), loss frequency (%), spread (bp), printed then exact; $|z|$ in standard errors at 20,000 draws

T	Cell	No-collapse		Loss		Spread		$\|z\|$
9	High / A	–	–	–	–	2.00	1.03	1.35
9	High / AA	–	–	–	–	0.00	0.34	0.82
9	Very high / NR	–	–	–	–	4.50	7.58	1.58
9	Very high / B	–	–	–	–	6.00	5.75	0.15
9	Very high / A	–	–	–	–	3.50	2.93	0.47
9	Very high / AA	–	–	–	–	1.00	1.03	0.04
10	2 / NR	–	–	–	–	5.00	8.16	1.57
10	2 / B	–	–	–	–	4.00	6.52	1.40
10	2 / A	–	–	–	–	2.00	3.71	1.26
10	2 / AA	–	–	–	–	0.50	1.49	1.15
10	10 / NR	–	–	–	–	1.50	2.07	0.56
10	10 / B	–	–	–	–	0.50	1.57	1.21
10	10 / A	–	–	–	–	0.00	0.80	1.27
10	10 / AA	–	–	–	–	0.00	0.27	0.74
10	20 / NR	–	–	–	–	0.00	0.67	1.15
10	20 / B	–	–	–	–	0.00	0.49	0.99
10	20 / A	–	–	–	–	0.00	0.23	0.67
10	20 / AA	–	–	–	–	0.00	0.07	0.37
10	50 / NR	–	–	–	–	0.00	0.11	0.47
10	50 / B	–	–	–	–	0.00	0.08	0.39
10	50 / A	–	–	–	–	0.00	0.03	0.25
10	50 / AA	–	–	–	–	0.00	0.01	0.12

6 Model Limitations and Known Weaknesses

1. **Benchmark, not validation.** The credit is shown to be computed as documented; the seeds remain uncalibrated (SEIS-R1, R2).

2. **Precision.** Price the spread from the exact credit, or raise the draw count until its standard error is below the precision quoted, and print the sensitivity tables from the exact credit so their orderings mean something.

3. **Documentation.** State the GMPE's functional form and coefficients (§3.3, Appendix B); call the Table 18 figures medians of clipped lognormals, or set them so the means are as stated; withdraw §9.6's stability claim.

7 Document History

Date	Version	Description	Author
28-Oct-2026	0.1	First issue	David K Kelly